PAYEL BHATTACHARJEE

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RESEARCH SUMMARY

Ph.D. candidate at University of Arizona specializing in efficient and trustworthy machine learning, differential privacy, large language models (LLMs), generative AI, alignment and reward modeling, causal inference and causal graph discovery (CGD) and scalable deep learning systems. Experience in LLM alignment (RLHF), reward modeling, and fine-tuning, with a focus on building and evaluating models on large-scale real-world datasets and benchmarks.

EXPERIENCE

University of Arizona — Tucson, AZ
Graduate Research Assistant — *Aug 2022 – Present*

- Fine-tuned transformer-based models for LLM alignment using HH-RLHF, UltraFeedback, PKU-SafeRLHF datasets.
- Designed adaptive privacy-preserving LLM classifier and causal graph discovery algorithm.
- Designed framework for hard-sample based reward modeling, improving model robustness on low-margin preferences.
- Built scalable training pipelines (PyTorch, HuggingFace, LoRA) on GPU clusters and HPC environments for large-scale experiments and data analysis.
- Evaluated model performance using RewardBench and AlpacaEval-style benchmarks; trained and fine-tuned LLaMA, RoBERTa, DeBERTa, and BERT models for alignment and text classification tasks.

Bosch Global Software Technologies (BGSW) — India
Associate Software Engineer — *Jul 2021 – Jun 2022*

- Developed ADAS software solutions and automation scripts for automotive clients, streamlining data analysis workflows across the U.S. and Europe and improving efficiency through machine learning and driven data processing.

Project Intern — *Mar 2021 – Jun 2021*

- Developed machine learning frameworks and Python solutions to streamline workflows, accelerate data analysis, and optimize model performance.

University of Arizona — Tucson, AZ
Graduate Teaching Assistant — *Aug 2022 – May 2023*

- Teaching Assistant for ECE 372, supporting 60 students with laboratory instruction, code debugging, and hands-on guidance while coordinating lab activities to ensure effective learning.

EDUCATION

Ph.D., Electrical and Computer Engineering — Aug 2022 – Present
University of Arizona (Tucson, AZ, USA) — GPA: 3.79/4.00

B.Tech., Electronics & Telecommunication Engineering — 2017 – 2021
Kalinga Institute of Industrial Technology (KIIT) (Bhubaneswar, Odisha, India) — GPA: 9.43/10.0

TECHNICAL SKILLS

LLMs & GenAI: Transformers, RLHF, speculative decoding, federated learning, split learning, conformal prediction.
Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA.
NLP: Text classification, QA-based performance analysis, evaluation pipelines.
Programming & Systems: Python, C/C++, R, Matlab, GPU clusters, HPC environments, multi-GPU training.
Trustworthy ML: Alignment, reward modeling, differential privacy, causal inference, causal graph discovery.

AWARDS & SERVICES

UofA GPSC Travel Grants (2024, 2025)
People's Choice Award, ECE Graduate Poster Symposium (2024)
KIIT Merit Scholarship (2018, for securing 10.0/10.0 CGPA & SGPA).
UofA ECE Graduate Student Association (GSA) officer (2024 - present).

PUBLICATIONS

AI Alignment & Efficiency

1. **P. Bhattacharjee**, et al. *MARS: Margin and Semantic-Aware Data Augmentation for Reward Modeling*, *arXiv preprint*, 2026 (Under Review).
2. N. Teku, F. Tian, **P. Bhattacharjee**, et al. *PROPS: Progressively Private Self-alignment of LLMs*, *TMLR*, 2025.
3. **P. Bhattacharjee***, F. Tian*, M. Zhong*, et al. *Conformal Sparsification for Bandwidth-Efficient Speculative Decoding*, *NeurIPS AI4NextG*, 2025.

Privacy Preserving AI

4. **P. Bhattacharjee**, et al., *Learning to Diagnose Privately: DP-Powered LLMs for Radiology Report Classification*, *IEEE Access*, 2026.
5. F. Tian, **P. Bhattacharjee**, et al. *STAMP: Selective Task-Aware Mechanism for Text Privacy*, *EACL*, 2026.
6. **P. Bhattacharjee**, R. Tandon., *CURATE: Scaling Up Differentially Private Causal Graph Discovery*, *Entropy*, 2024.
7. **P. Bhattacharjee**, R. Tandon, *Adaptive Privacy for Differentially Private Causal Graph Discovery*, *IEEE MLSP*, 2024.
8. Z. Guan, Z. Zhao, F. Tian, D. Nguyen, **P. Bhattacharjee**, et al., *A Framework for Multi-source Privacy Preserving Epidemic Analysis*, *arXiv preprint*, 2025.

Miscellaneous

9. **P. Bhattacharjee**, et al., *Sleep and Sedentary Behavior Analysis from Physiological Signals Using Machine Learning*, *IEEE ICIMIA*, 2020.
10. **P. Bhattacharjee**, et al., *Numerical Modelling and Performance Evaluation of SnS-Based Heterojunction Solar Cell with p+-SnS BSF Layer*, *Optical and Quantum Electronics*, 2022.